

The Hong Kong Polytechnic University

Department of Computing

COMP4913 Capstone Project

Meeting Note

**Food Component Detection for Dietary
Recommendation**

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COMP4913 Capstone Project Meeting Notes

Student Name: LIU Yibo

Student ID: 22098545D

Project Title: Food Component Detection for Dietary Recommendation - Children's Fun Nutrition Guide

Supervisor: Dr PANG WaiMan

Meeting Note #1

Date: September 4, 2025

Time: 15:00-16:00

Location/Platform: PQ821

Meeting Agenda

- Initial project brainstorming session
- Discussion of overall project theme: Dietary Component Detection and Recommendation
- Clarification of project structure and requirements

Discussion Summary

The supervisor initiated a comprehensive discussion regarding the overarching theme of dietary component detection and recommendation systems. A critical clarification was made that **this would be an individual project**, requiring each student to work independently despite sharing a similar general theme with other students in the group.

Dr. Pang emphasized that while the broad topic area would be consistent across team members, each student must develop a **unique and distinct project** with different focus areas, target users, or implementation approaches. This ensures that each student's work represents independent effort and original contribution.

The meeting concluded with a collaborative brainstorming session where all participants shared preliminary ideas and potential directions for their individual projects. Diverse perspectives helped identify various possibilities within the dietary recommendation domain.

Supervisor's Feedback

- Individual project requirement clearly communicated
- Need for differentiation between each student's project emphasized
- Encouraged creative thinking and exploration of various angles within the dietary recommendation theme

Action Points

- Conduct independent research on potential project angles and target user groups
- Brainstorm specific, unique project ideas that align with personal interests and technical capabilities
- Prepare preliminary project concepts for discussion at next meeting
- Review existing dietary recommendation systems and identify gaps or opportunities

Problems Encountered

- Initial uncertainty about project scope and individualization requirements
- Need to distinguish individual projects while maintaining thematic consistency

Meeting Note #2

Date: September 17, 2025

Time: 15:00-16:00

Location/Platform: PQ821

Meeting Agenda

- Review and evaluate preliminary project concept
- Discuss the guest speaker's recommendations
- Refine project focus and target audience
- Initial technical approach discussion

Discussion Summary

Building upon the previous meeting's brainstorming session, this meeting focused on **in-depth analysis and refinement** of my preliminary project concept. I presented my initial ideas regarding a dietary recommendation system, which underwent critical evaluation for feasibility and scope.

The supervisor arranged for valuable input from **Dr. KC Kwan, Assistant Professor from the Computer Science Department at California State University, Sacramento, USA**, who provided an expert perspective on dietary recommendation systems and health-focused applications. Dr. Kwan's insights, combined with Dr. Pang's guidance, significantly shaped the project direction.

Key recommendations received:

1. **Narrow the target audience** to a specific user demographic rather than attempting a general-purpose solution
2. **Choose a specific platform** - either a web application or mobile app, with clear

justification

3. **Define a concrete, consistent database structure** for food and nutritional information
4. **Focus on user experience** tailored to the chosen demographic

The discussion emphasized the importance of **specificity and focus** on capstone projects, avoiding overly broad scopes that would be difficult to implement within the project timeline.

Supervisor's Feedback

- Current concept is too broad and needs a more specific focus
- The target audience must be clearly defined with a specific age range and characteristics
- Platform selection (web vs. mobile) should be decided based on target user needs
- Database design should be planned early to ensure consistency throughout development
- Consider existing solutions and identify a unique value proposition

Action Points

- Research existing dietary recommendation applications and identify target audience gaps
- Decide between web applications (WordPress-based) or mobile app development
- Define a specific target user group with age range and characteristics
- Draft initial database scheme for food and nutritional information (minimum 100 food items)
- Prepare a refined project proposal incorporating feedback from both supervisors
- Identify potential data sources for nutritional information (WHO, CDC guidelines)

Problems Encountered

- Initial project scope was too ambitious and unfocused
- Difficulty in choosing between multiple potential target audiences

- Uncertainty about technical platform selection and its implications

Solutions Proposed

- Focus on children aged 3-12 and their parents as primary target users
- Select WordPress for web development due to its extensive plugin ecosystem and rapid development capability
- Incorporate gamification elements to engage young users while providing scientific nutrition guidance for parents

Meeting Note #3

Date: October 9, 2025

Time: 15:00-16:00

Location/Platform: PQ821

Meeting Agenda

- Finalize project scope and objectives
- Confirm target audience and platform selection
- Discuss core features and functionality
- Review project feasibility and timeline

Discussion Summary

This meeting marked a significant milestone in finalizing my project direction. After incorporating feedback from previous meetings and conducting extensive research, I presented a **comprehensive and focused project concept** that received supervisor approval.

Finalized Project Concept:

Project Title: Children's Fun Nutrition Guide

Platform: WordPress-based web application

Website: <https://kids-nutrition-fun.com> (domain registered and hosting established)

Target Users: Children aged 3-12 years old and their parents

Core Problem Statement: The project addresses a critical gap in the current market - the **lack of engaging, personalized digital tools** that effectively guide parents and children aged 3-12 toward balanced nutrition. Existing platforms often fail to incorporate interactive elements (such as games or animated recipes) that can motivate children to develop healthy habits, particularly in areas of:

- Growth and development support
- Immune system enhancement
- Overcoming picky eating behaviors

Project Approach: Develop a user-friendly web system using WordPress that combines:

1. **Personalized food recommendations** based on the child's age, health goals, and dietary restrictions
2. **Interactive gamification elements** to engage children in learning about nutrition
3. **Evidence-based content** derived from WHO and CDC guidelines
4. **Parent-friendly features**, including progress tracking and nutritional reports

The supervisor emphasized that the project's strength lies in its **dual-audience approach** - simultaneously engaging children through fun, interactive elements while providing parents with credible, actionable nutritional guidance.

Supervisor's Feedback

- Excellent refinement from initial concept - clear focus and well-defined scope
- Target audience (3-12 age range with parents) is appropriate and manageable
- WordPress selection justified given rapid development needs and plugin ecosystem
- Strong foundation in authoritative sources (WHO, CDC) adds credibility
- Gamification approach aligns well with the target audience engagement needs
- Domain registration and hosting setup demonstrate proactive project management

Recommendations Provided

- Ensure frontend interaction design is truly child-friendly (large buttons, cartoon themes,

simple navigation)

- Focus recommendation logic on practical, achievable meal planning rather than complex algorithms
- Balance scientific accuracy with age-appropriate presentation
- Consider privacy and security carefully, given child data protection requirements
- Plan database structure to support at least 100 child-friendly foods with comprehensive nutritional data

Action Points

- Register domain and establish hosting (completed with SiteGround)
- Set up WordPress development environment
- Begin detailed functional requirements documentation
- Design a user registration and profile management system for multiple children
- Research and evaluate WordPress plugins for user management, forms, and gamification
- Create initial database schema for food and nutritional information
- Draft project proposal document incorporating all finalized details
- Prepare a literature review on child nutrition guidelines and digital health tools

Problems Encountered

- Need to balance engaging design for children with professional credibility for parents
- Technical uncertainty about implementing personalized recommendation logic in WordPress
- Concerns about data privacy and security for child-related information

Solutions Proposed

- Use separate UI themes/sections for child-facing and parent-facing features
- Implement a recommendation system using WordPress custom post types and

taxonomies with custom shortcodes

- Utilize SiteGround's built-in security features (HTTPS, SSL) and implement role-based access control
- Research WCAG compliance for accessibility

Meeting Note #4

Date: October 23, 2025

Time: 15:00-16:00

Location/Platform: PQ821

Meeting Agenda

- Review complete draft of project proposal
- Discuss functional and non-functional requirements
- Evaluate the literature review comprehensiveness
- Refine methodology and project schedule
- Address resource estimation and budget

Discussion Summary

This meeting centered on a **comprehensive review of my draft project proposal document**. I submitted the complete draft covering all required sections: background and problem statement, objectives and outcomes, methodology, project schedule, resource estimation, and references.

Dr. Pang conducted a thorough review of the document, providing detailed feedback on multiple sections to enhance clarity, feasibility, and academic rigor.

Detailed Feedback by Section

1. Functional Requirements:

- Some proposed features needed clarification on the implementation approach
- Recommendation to specify the exact number of recipe options per health goal (changed to 3-5 customized options)

- Suggestion to clarify interactive mini-game concepts and their educational objectives
- Need to specify downloadable report formats and content structure

2. Non-Functional Requirements:

- Performance metrics needed more specific quantification
- Added concrete targets: page load <3s, recommendation generation <5s, support for 10 concurrent users
- Security requirements strengthened with explicit mention of HTTPS, data encryption, and role-based access control
- Usability standards refined to include WCAG compliance and touch-friendly interface specifications

3. Literature Review:

- Supervisor commended the inclusion of authoritative sources (WHO, CDC, UNICEF)
- Recommended adding more recent research articles on digital health interventions for children
- Suggested strengthening the connection between literature findings and project justification
- Advised clarifying how existing research gaps inform the project's unique contribution

4. Methodology:

- Overall approach approved as systematic and well-structured
- Recommended breaking down the Implementation phase into more granular tasks
- Suggested adding specific testing metrics and user feedback collection methods
- Advised on including contingency plans for potential technical challenges

5. Project Schedule:

- Timeline assessed as realistic and achievable
- Minor adjustments suggested for Testing & Optimization phase to allow adequate iteration time

- Recommended building buffer time for unexpected delays
- Emphasized the importance of consistent GitHub commits for progress tracking

6. Resource Estimation:

- Hosting choice (SiteGround) justified and appropriate
- Budget for Azure OpenAI API approved for AI-assisted development and research
- Advised maintaining detailed records of plugin selections and costs as the project progresses
- Suggested preparing backup options if primary tools encounter limitations

Supervisor's Feedback Summary

Strengths Identified:

- Well-researched background and clear problem statement
- Comprehensive coverage of both functional and non-functional requirements
- Realistic resource planning with justified selections
- Strong foundation in evidence-based nutritional guidelines
- Proactive infrastructure setup (domain, hosting, repository)

Areas for Improvement:

- Enhance specificity in functional requirement descriptions
- Strengthen literature review with additional recent academic sources
- Clarify technical implementation details for recommendation logic
- Expand on user testing methodology and feedback collection approach
- Ensure all requirements are measurable and testable

Action Points

- Revise functional requirements with more specific implementation details
- Add concrete performance metrics to non-functional requirements

- Expand literature review with 2-3 additional recent research articles
- Clarify the recommendation algorithm approach in the methodology section
- Specify user testing protocol with sample size and feedback mechanisms
- Include data privacy and security compliance details
- Finalize project proposal with all supervisors' recommendations incorporated
- Submit final project proposal by October 24, 2025, deadline
- Begin Phase 1 implementation immediately after proposal submission

Problems Encountered

- Initial draft lacked sufficient technical detail in some sections
- Need to balance comprehensive documentation with concise presentation
- Uncertainty about exact plugin selections requiring "TBD" budget items
- Challenge in estimating accurate time requirements for unfamiliar WordPress development tasks

Solutions Implemented

- Added specific metrics and quantifiable targets throughout the document
- Structured technical sections with clear subsections for readability
- Created provisional budget category for WordPress plugins with a plan to update as decisions are made
- Consulted WordPress development resources and tutorials to inform realistic time estimates
- Added contingency buffer to schedule for learning curve and unexpected challenges

Project Status

- **Domain and Hosting:** Established (<https://kids-nutrition-fun.com>)
- **GitHub Repository:** Created and configured
- **Development Environment:** WordPress installed and configured

- **Proposal Document:** Draft complete, awaiting final revisions

Meeting Note #5

Date: January 5, 2026

Time: 14:00-15:00

Location/Platform: Online (Zoom)

Meeting Agenda

- Review overall project progress and completion status
- Demonstrate current website functionality
- Discuss interim report preparation
- Plan remaining development phases (February-April 2026)
- Address technical challenges and solutions

Discussion Summary

This meeting served as a **comprehensive progress review** before the interim report submission deadline (January 9, 2026). I presented the live website at <https://kids-nutrition-fun.com>, demonstrating completed functionality and discussing the development journey from October through early January.

Current Project Status:

The project has achieved approximately **50% completion** with substantial progress across all foundational areas. The website is fully operational with core infrastructure established, including:

- Complete technical setup: SiteGround hosting (Singapore), WordPress 6.x with SSL (A+ security rating)
- Fully responsive UI design with child-friendly interface using Elementor
- Six complete pages: Home, About, Recipes, Game, Contact, and Individual Recipe pages
- Prototype food database with 9 recipes categorized by meal type (3 breakfast, 3 lunch,

3 dinner)

- Functional recipe filtering and browsing system using WooCommerce
- Email inquiry system with pre-filled templates
- Interactive educational game integration (USDA MyPlate)
- GitHub repository with regular commits

Live Demonstration:

I walked Dr. Pang through the website, showcasing:

1. Homepage design with dual-audience approach (parent-professional, child-playful)
2. Recipe catalog with category filtering functionality
3. Educational game section with clear learning objectives
4. Contact system enabling parent inquiries
5. Mobile responsiveness across devices

Supervisor's Feedback

Strengths Identified:

- Excellent progress with solid technical foundation established
- Effective dual-design strategy balancing parent credibility and child engagement
- High-quality UI implementation with professional appearance
- Successful integration of external educational resources
- Comprehensive testing methodology demonstrating quality assurance
- Clear documentation of challenges encountered and solutions applied
- Realistic timeline with achievable milestones

Positive Assessment:

Dr. Pang expressed satisfaction with the **current progress trajectory**, noting that the 50% completion status aligns appropriately with the project timeline. The functioning website

demonstrates practical implementation skills and effective project management.

Recommendations for Remaining Development

Priority Features to Implement:

1. User Registration and Authentication System

- Implement Ultimate Member plugin for parent account creation
- Design multi-child profile management dashboard
- Configure role-based access control for data privacy
- Include email verification and secure password requirements

2. Recipe Database Expansion

- Scale from 9 prototype recipes to **100+ comprehensive entries**
- Add complete ingredient lists and step-by-step instructions
- Include detailed nutritional information panels (calories, macros, micronutrients)
- Implement recipe search functionality by ingredients, allergens, and health goals

3. AI Chatbot Integration (Stretch Goal)

- Integrate Azure OpenAI API for FAQ assistance if time permits
- Provide recipe suggestions and troubleshooting support
- Create rule-based fallback if AI implementation exceeds timeline

4. Nutrition Reports and Progress Tracking

- Design downloadable PDF reports with visual charts
- Develop progress dashboards showing food group diversity
- Implement weekly/monthly summary generation

Action Points

Immediate (January 2026):

- Complete and submit interim report by January 9, 2026

- Document current implementation details, testing results, and challenges
- Include comprehensive screenshots and architecture diagrams
- Outline detailed roadmap for remaining development phases

February 2026 - Implementation Completion:

- Implement user registration and multi-child profile management
- Expand recipe database to 100+ entries with complete nutritional data
- Develop personalized recommendation algorithm
- Enhance gamification features with custom games

March 2026 - Testing & Optimization:

- Conduct comprehensive user testing with 8-12 parent-child pairs
- Optimize performance (image compression, caching, database indexing)
- Address all identified bugs and accessibility issues
- Perform security hardening and compliance verification

April 2026 - Documentation & Presentation:

- Implement nutrition reports if time permits
- Integrate AI chatbot as stretch goal
- Complete final project report with methodology and results
- Prepare demonstration video and presentation materials

Maintain flexibility to defer non-essential features to post-graduation enhancements.

Personal Reflection

This meeting provided valuable validation that the project is on track despite the substantial work remaining. The supervisor's positive feedback on current progress boosts confidence, while the clear prioritization of remaining features provides a focused roadmap. The emphasis on quality over quantity helps manage scope and reduces pressure to implement every

conceivable feature. I feel prepared to document current progress in the interim report and execute the remaining development phases systematically over the next three months.

Meeting Note #6

Date: January 20, 2026

Time: 14:00–15:00

Location/Platform: PQ821

Meeting Agenda

- Review interim report feedback from supervisor
- Discuss post-interim development priorities
- Address evaluation criteria and grading rubric
- Plan feature completion schedule for February–April 2026

Discussion Summary

This meeting took place shortly after the interim report submission (January 9, 2026) and focused primarily on reviewing Dr. Pang's written feedback on the interim report. The supervisor provided a structured evaluation of the submitted document, acknowledging the overall quality of the work while identifying key areas requiring further development before the final submission.

Dr. Pang confirmed that the project is progressing on schedule and that the website's current functionality demonstrates a solid technical foundation. Emphasis was placed on the importance of completing the remaining core features — particularly the user authentication system and recipe database expansion — before moving on to stretch goals such as the AI chatbot and downloadable nutrition reports.

Supervisor's Feedback

- Interim report is well-structured with a clear problem statement and project rationale

- Website design is child-friendly and visually appropriate for the target audience
- Recommendation to prioritise user login system and profile management as the next milestone
- Recipe database must be expanded to a minimum of 30 entries before the final submission, with nutritional data included
- Final report should include a dedicated evaluation section with quantitative user testing results
- Advised maintaining consistent GitHub commit history to demonstrate ongoing development progress

Problems Encountered

- Integrating the Ultimate Member plugin with WooCommerce for recipe access control presented unexpected compatibility issues
- Nutritional data sourcing for 100+ recipes is time-consuming; considering limiting to 30–50 high-quality entries

Project Status

- Website: Live and fully operational at <https://kids-nutrition-fun.com>
- Recipe database: 10 entries published, targeting 30+ by end of February
- User authentication: In planning phase
- AI chatbot (Nutri): Development to commence in February 2026

Meeting Note #7

Date: March 12, 2026

Time: 14:00–15:00

Location/Platform: PQ821

Meeting Agenda

- Demonstrate completed system features: AI chatbot, login system, and expanded recipe database
- Review overall system integration and functionality
- Discuss user survey results and feedback analysis
- Finalise development scope for remaining weeks

Discussion Summary

This meeting served as a mid-final-phase review, providing an opportunity to demonstrate the substantial system enhancements completed since January. I presented the live website with all major new features integrated, including the AI-powered Nutri Assistant chatbot, the user login and enrolment system, and the significantly expanded recipe database.

The AI chatbot integration was a key highlight of this meeting. The Nutri Assistant, powered by the COMP OpenAI API, is now fully functional and accessible via a floating widget on all pages. The chatbot can respond to nutrition-related queries in both English and Chinese, provide recipe suggestions based on the child's age and health goals, and assist parents with general dietary guidance.

The recipe database has been expanded to over 20 entries, with recipes sourced from Kids Cook Monday (mondaycampaigns.org) and organised into multiple categories including Breakfast & Brunch, Pasta & Grains, Veggie Mains, Salads & Bowls, Soups & Stews, Snacks & Treats, and Dips & Sauces. Each recipe now includes a full ingredient list, step-by-step instructions, and a nutritional information panel.

The user enrolment system has been implemented, allowing parents to register accounts and access personalised features. A brief discussion was held regarding the results of the preliminary user survey, which was distributed to a group of teachers and parents via a class group chat. Initial responses indicated high satisfaction with the website's fun and educational value.

Problems Encountered

- Several recipes were found to be in "Draft" status and therefore not visible to logged-out users — this explains the discrepancy between admin and visitor recipe counts observed earlier
- The AI chatbot's response latency can occasionally be noticeable on slower connections; currently relying on the API's default response time

Solutions Implemented

- All draft recipes have been reviewed and published; SiteGround cache cleared to ensure changes are visible to all visitors
- Added a loading indicator to the chatbot widget to manage user expectations during API response delays

Project Status

- Website: Fully operational with all core features live
- Recipe database: 20+ entries published across 6 categories
- AI chatbot (Nutri): Live and functional on all pages
- User authentication: Enrolment system active
- User survey: Distributed; responses being collected
- Final report: Drafting commenced

Meeting Note #8

Date: April 2, 2026

Time: 14:00–15:00

Location/Platform: PQ821

Meeting Agenda

- Final project review and overall assessment
- Review and refine final report structure
- Discuss user survey analysis and evaluation findings
- Plan final presentation (PPT) structure and delivery strategy
- Address any outstanding issues before submission

Discussion Summary

This final meeting served as a comprehensive review of the completed project ahead of the final report submission and capstone presentation. I presented the fully completed website at <https://kids-nutrition-fun.com>, demonstrating all implemented features and walking Dr. Pang through the key deliverables outlined in the original project proposal.

The supervisor conducted a thorough review of the draft final report, providing targeted feedback on structure, academic writing quality, and the depth of the evaluation section. Particular attention was given to ensuring that the report accurately reflects the technical implementation decisions made throughout the project, and that the evaluation findings are presented with sufficient analytical depth.

User survey results were reviewed and discussed. The survey, distributed to parents, teachers, and students via a group chat, collected responses evaluating the website's fun and engagement, learning value, educational merit, and overall satisfaction. Results were broadly positive, with participants rating the website highly for child-friendliness and educational content. Constructive feedback identified opportunities to add more interactive features and expand the recipe collection further.

Supervisor's Feedback

- Project has successfully met all core objectives defined in the proposal
- Website demonstrates strong technical implementation with a professional and child-appropriate design
- AI chatbot integration (Nutri) is a standout feature that differentiates the project from typical capstone submissions

- Final report requires stronger analysis in the evaluation section — move beyond summarising survey results to critically interpreting findings
- Recommended adding a "Limitations and Future Work" subsection to demonstrate academic reflection
- PPT presentation should follow a clear narrative arc: problem → solution → implementation → evaluation → reflection
- Advised rehearsing the live website demo at least twice before the presentation day to ensure smooth delivery
- Ensure all references in the report follow the required citation format consistently

Action Points

- Finalise and proofread the complete final report before submission deadline
- Strengthen the evaluation section with critical analysis of user survey results
- Rehearse the live website demonstration and ensure stable internet connection on presentation day
- Submit final report and all supporting materials by the official deadline
- Prepare a 2–3 minute project demonstration video as a backup for the live demo

Final Personal reflections

This final meeting marks the conclusion of a rewarding and challenging capstone project. From the initial brainstorming sessions in September 2025 to a fully functioning, AI-powered nutrition website for children, the project has grown considerably in scope and technical ambition. The experience of designing, building, and evaluating a real-world web application — from domain registration to AI chatbot integration — has significantly deepened my understanding of full-stack web development, user-centred design, and evidence-based

health information systems. I am grateful for Dr. Pang's consistent guidance and feedback throughout the project, which played a critical role in keeping the work focused and academically rigorous.